

# Notes: Kedia, Rajgopal & Zhou (JFE, 2014)

## Did going public impair Moody's credit ratings?

### Contents

|          |                                                                                        |           |
|----------|----------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Big picture</b>                                                                     | <b>3</b>  |
| <b>2</b> | <b>Data and measurement</b>                                                            | <b>4</b>  |
| 2.1      | Why Moody's IPO is useful . . . . .                                                    | 4         |
| 2.2      | Corporate bond issue sample . . . . .                                                  | 4         |
| 2.3      | Rating scale and RatingDiff . . . . .                                                  | 4         |
| 2.4      | Issuer and issue controls . . . . .                                                    | 4         |
| 2.5      | Conflict-of-interest proxies . . . . .                                                 | 5         |
| 2.6      | Outstanding-bond data . . . . .                                                        | 5         |
| 2.7      | Informativeness and default data . . . . .                                             | 5         |
| 2.8      | Structured-product data . . . . .                                                      | 6         |
| <b>3</b> | <b>Models and empirical specifications</b>                                             | <b>7</b>  |
| 3.1      | Difference-in-differences logic . . . . .                                              | 7         |
| 3.2      | Baseline new-issue regression . . . . .                                                | 7         |
| 3.3      | Cross-sectional conflict-of-interest model . . . . .                                   | 7         |
| 3.4      | Outstanding-bond model . . . . .                                                       | 7         |
| 3.5      | Informativeness model . . . . .                                                        | 8         |
| 3.6      | Structured-product rating tests . . . . .                                              | 8         |
| 3.7      | Cross-sectional intercept tests . . . . .                                              | 8         |
| <b>4</b> | <b>Identification and empirical design</b>                                             | <b>9</b>  |
| 4.1      | What identifies the main effect . . . . .                                              | 9         |
| 4.2      | Why S&P is the benchmark . . . . .                                                     | 9         |
| 4.3      | Addressing issuer composition . . . . .                                                | 9         |
| 4.4      | Addressing S&P conservatism . . . . .                                                  | 9         |
| 4.5      | Why conflict proxies matter . . . . .                                                  | 9         |
| 4.6      | Why informativeness tests matter . . . . .                                             | 10        |
| 4.7      | What the paper cannot fully prove . . . . .                                            | 10        |
| <b>5</b> | <b>Tables and figure</b>                                                               | <b>11</b> |
| 5.1      | Table 1 + Table 2: rating scale and sample descriptives . . . . .                      | 11        |
| 5.2      | Table 3 + Figure 1: the basic pre/post shift . . . . .                                 | 12        |
| 5.3      | Table 4 + Table 5: main new-issue regression and structured-product issuance . . . . . | 12        |
| 5.4      | Table 6 + Table 7: cross-sectional conflicts and outstanding bonds . . . . .           | 13        |
| 5.5      | Table 8 + Table 9 + Table 10: informativeness and alternative explanations . . . . .   | 13        |
| 5.6      | Table 11 + Table 12: structured products and CDO ratings . . . . .                     | 15        |
| 5.7      | Table 13 + Table 14: robustness and cross-sectional tests . . . . .                    | 15        |
| <b>6</b> | <b>Short summary</b>                                                                   | <b>16</b> |
| <b>7</b> | <b>Conclusion</b>                                                                      | <b>17</b> |

|     |                              |    |
|-----|------------------------------|----|
| 7.1 | Core conclusion . . . . .    | 17 |
| 7.2 | Economic intuition . . . . . | 17 |
| 7.3 | Takeaway . . . . .           | 17 |

# 1 Big picture

- **Question.** Did Moody's public listing in 2000 weaken its credit rating standards by increasing pressure to grow revenue and market share?
- **Allegation.** Congressional testimony after the financial crisis claimed that Moody's shifted from a reputation-oriented culture to a revenue-focused culture after becoming an independent publicly traded company.
- **Empirical idea.** The paper benchmarks Moody's ratings against S&P ratings on the same securities before and after Moody's IPO. S&P did not experience a comparable ownership change at the time.
- **Main sample.** The core corporate bond sample contains 8,505 new issues rated by both Moody's and S&P from 1995–2005, excluding 2000.
- **Key variable.**  $RatingDiff = \text{S\&P numerical rating} - \text{Moody's numerical rating}$ . Since lower numerical values correspond to better ratings, a positive  $RatingDiff$  means Moody's gives the more favorable rating.
- **Main result.** Moody's was tougher than S&P before the IPO and more favorable after the IPO. The average  $RatingDiff$  rises from  $-0.302$  before the IPO to  $0.286$  after it, a shift of about 0.59 rating notches.
- **Economic magnitude.** The paper translates the 0.588-notch increase into roughly a 13 basis point reduction in yield, a 10 basis point lower implied default rate, and a 5.5 basis point lower expected credit loss for the average bond.
- **Client-conflict evidence.** Moody's post-IPO relative rating generosity is stronger for firms with greater conflict-of-interest or market-share relevance: structured-product issuers, financial firms, large/frequent bond issuers, marginal issuers that narrowly miss a higher S&P rating, and industries with greater Fitch competition.
- **Outstanding-bond evidence.** The same pattern appears for outstanding bonds, not just newly issued bonds.
- **Informativeness test.** The more favorable post-IPO Moody's ratings are not accompanied by lower expected default frequencies or lower observed defaults.
- **Alternative explanations.** The paper finds little support for explanations based on higher recovery rates, S&P conservatism, or Fitch's expansion alone.
- **Structured products.** In a CDO sample, Moody's also gives relatively more favorable ratings than S&P after its IPO.
- **Policy takeaway.** Public ownership can sharpen market and revenue incentives for financial gatekeepers, potentially weakening professional standards when quality is difficult to measure in real time.

## 2 Data and measurement

### 2.1 Why Moody's IPO is useful

- Moody's was spun off from Dun & Bradstreet and became a separately traded company on October 4, 2000.
- The research design treats this ownership change as a shock to Moody's incentives.
- The pre-public period is 1995–1999; the post-public period is 2001–2005; 2000 is excluded.
- S&P is the benchmark agency because it did not have a comparable ownership change in the event window.

**Interpretation:** The empirical design is not a pure randomized experiment. It is a difference-in-differences style comparison of Moody's ratings relative to S&P's ratings on the same bond issues, before and after the Moody's IPO.

### 2.2 Corporate bond issue sample

- New corporate bond data come from Mergent's Fixed Income Securities Database (FISD).
- The sample includes new U.S. corporate bond issues rated by both Moody's and S&P.
- Government agency bonds, Freddie Mac, Fannie Mae, bonds rated AAA by both agencies, and bonds with rating disagreements of four or more notches are excluded.
- The final new-issue sample has 8,505 bonds: 5,722 before the IPO and 2,783 after the IPO.

**Interpretation:** The requirement that both agencies rate the same issue is central. It allows ratings to be compared security by security, reducing selection and issuer-composition concerns.

### 2.3 Rating scale and RatingDiff

- Moody's and S&P letter ratings are converted into a numerical scale from 1 to 21.
- Lower values mean better ratings: Aaa/AAA equals 1, Aa1/AA+ equals 2, and so on down to C.
- For each issue  $i$ , the paper defines

$$RatingDiff_i = Rating_i^{S\&P} - Rating_i^{Moody's}.$$

- $RatingDiff_i > 0$  means Moody's gives a more favorable rating than S&P.
- $RatingDiff_i < 0$  means Moody's gives a tougher rating than S&P.

**Interpretation:** This definition is easy to misread because smaller numerical ratings are better. The sign of  $RatingDiff$  is chosen so that a positive coefficient means relative Moody's laxity.

### 2.4 Issuer and issue controls

The main regressions control for bond and issuer characteristics that can affect ratings:

- issuer size: log market value of equity plus book value of debt;
- leverage: long-term debt divided by total assets;
- operating margin: operating income before depreciation divided by sales;
- stock return standard deviation in the year before issuance;

- issue size: log par value of the bond;
- years to maturity;
- senior-debt indicator.

**Interpretation:** These controls address the possibility that Moody’s and S&P rate different kinds of issuers differently and that issuer quality or bond characteristics changed after 2000.

## 2.5 Conflict-of-interest proxies

The paper uses five proxies for settings in which Moody’s had stronger incentives to cater to clients after becoming public:

- **Structured-product issuers:** firms among the top 40 issuers of ABS, MBS, and CDO products in the prior year.
- **Financial firms:** firms in banking, credit, finance, and savings/lending industries.
- **Large/frequent corporate bond issuers:** issuers above the median in par value of bonds issued over the prior three years.
- **Marginal issuers:** issuers with high operating profits relative to others in the same S&P rating class, making a higher Moody’s rating especially valuable.
- **High Fitch market-share industries:** industries where Fitch competition is above the median.

**Interpretation:** The conflict tests ask whether rating leniency is stronger where a public Moody’s had the most to gain from client relationships and market share.

## 2.6 Outstanding-bond data

- The paper extends the analysis beyond new issues to outstanding bonds.
- For outstanding bonds, ratings can change at different times and by different magnitudes across agencies.
- The paper constructs *LeadTimeDiff*: the fraction of a year in which Moody’s has a higher rating than S&P minus the fraction of a year in which S&P has a higher rating than Moody’s.

**Interpretation:** Outstanding-bond tests ask whether Moody’s post-IPO generosity affected continuing ratings, not merely new-issue ratings designed to win business.

## 2.7 Informativeness and default data

- Expected default frequencies (EDFs) are estimated using a Black–Scholes–Merton distance-to-default approach.
- Bond defaults are collected from FISD and NAIC bond transaction data.
- Recovery rates are estimated using market bid prices around default dates.

**Interpretation:** If Moody’s higher post-IPO ratings reflected superior information, they should predict lower default risk. The paper finds little evidence of this.

## 2.8 Structured-product data

- Structured-product issuer data come from J.P. Morgan's ABS database.
- CDO ratings are collected from Bloomberg for 1997–2003, excluding 2000.
- The CDO tests compare Moody's and S&P ratings at the deal and tranche level.

**Interpretation:** The structured-product analysis asks whether the same post-IPO rating tilt appears in the area most directly linked to the financial-crisis allegations.

### 3 Models and empirical specifications

#### 3.1 Difference-in-differences logic

The core empirical comparison is:

Moody's relative rating favorability after IPO – Moody's relative rating favorability before IPO.

Since *RatingDiff* benchmarks Moody's against S&P on the same security, the paper is effectively estimating whether Moody's changed relative to S&P after Moody's public listing.

**Interpretation:** The identifying logic is that S&P absorbs common rating-standard changes and bond-market trends, while the post-IPO indicator captures a Moody's-specific shift.

#### 3.2 Baseline new-issue regression

For new bond issue  $i$ , the baseline model is

$$RatingDiff_i = \gamma_0 + \gamma_1 post2000Dum_i + \sum_{j=2}^8 \gamma_j ControlVar_i^j + \sum_{j=2}^8 \delta_j \left( ControlVar_i^j \times post2000Dum_i \right) + \varepsilon_i. \quad (1)$$

- $post2000Dum_i = 1$  for 2001–2005 issues and zero for 1995–1999 issues.
- $\gamma_1 > 0$  implies Moody's becomes more favorable relative to S&P after the IPO.
- Interactions allow rating models to change after 2000.

**Interpretation:** The coefficient on  $post2000Dum$  is the main evidence on relative loosening. Table 4 estimates this coefficient at 0.585 without controls and 0.851 with controls.

#### 3.3 Cross-sectional conflict-of-interest model

The conflict tests estimate

$$RatingDiff_i = \gamma_0 + \gamma_1 post2000Dum_i + \gamma_2 HighConfDum_i + \gamma_3 (HighConfDum_i \times post2000Dum_i) + \Gamma' X_i + \Pi' (X_i \times post2000Dum_i) + \varepsilon_i. \quad (2)$$

- $HighConfDum_i$  equals one for a high-conflict client or setting.
- $\gamma_3 > 0$  means the post-IPO Moody's tilt is stronger for conflict-prone clients.

**Interpretation:** This specification asks whether the IPO effect is concentrated where revenue and market-share incentives are strongest.

#### 3.4 Outstanding-bond model

For outstanding bonds, the dependent variable becomes *LeadTimeDiff* <sub>$i$</sub> :

$$LeadTimeDiff_i = \gamma_0 + \gamma_1 post2000Dum_i + \gamma_2 HighConfDum_i + \gamma_3 (HighConfDum_i \times post2000Dum_i) + \Gamma' X_i + \varepsilon_i. \quad (3)$$

**Interpretation:** Positive coefficients mean Moody's spends a larger fraction of time above S&P in rating favorability. This test captures rating-maintenance behavior rather than only initial ratings.

### 3.5 Informativeness model

To test whether Moody’s higher ratings after the IPO are more informative, the paper estimates models such as

$$RatingDiff_i = \gamma_0 + \gamma_1 post2000Dum_i + \gamma_2 EDF_i + \gamma_3 (EDF_i \times post2000Dum_i) + \Gamma' X_i + \varepsilon_i, \quad (4)$$

or replaces  $EDF_i$  with a default indicator.

**Interpretation:** If Moody’s favorable post-IPO ratings reflected superior information, the interaction should indicate that Moody’s higher relative ratings are associated with lower expected or realized default risk. The paper finds little evidence for that.

### 3.6 Structured-product rating tests

For CDOs, the paper defines rating differences at two levels:

- **Deal level:** fraction of the deal rated AAA by Moody’s minus fraction rated AAA by S&P.
- **Tranche level:** an indicator that Moody’s rates the tranche AAA minus an indicator that S&P rates it AAA.

The same pre/post comparison is then applied to structured-product ratings.

**Interpretation:** Positive post-IPO differences imply that Moody’s assigned relatively more AAA treatment than S&P in CDOs after the IPO.

### 3.7 Cross-sectional intercept tests

For each year or quarter, the paper estimates

$$Y_i = \gamma_0 + \sum_{j=2}^8 \gamma_j ControlVar_i^j + \varepsilon_i, \quad (5)$$

where  $Y_i$  is *RatingDiff* for new issues or *LeadTimeDiff* for outstanding issues.

**Interpretation:** The intercept  $\gamma_0$  captures the average Moody’s-versus-S&P tilt after controls. Table 14 shows post-public intercepts are significantly higher than pre-public intercepts.

## 4 Identification and empirical design

### 4.1 What identifies the main effect

- The identifying variation is Moody’s ownership change in 2000.
- The comparison is Moody’s rating relative to S&P’s rating for the same security.
- The pre/post window excludes 2000 and compares 1995–1999 to 2001–2005.
- Controls and interactions address changing issuer and issue characteristics.

**Interpretation:** The design does not claim that Moody’s IPO was randomized. Its credibility comes from benchmarking to S&P on the same securities and from multiple robustness tests.

### 4.2 Why S&P is the benchmark

- S&P did not undergo a comparable ownership change in the sample window.
- Most U.S. corporate bonds are rated by both Moody’s and S&P.
- Comparing the two agencies on the same bond controls for bond-level fundamentals.

**Interpretation:** If both agencies loosened standards equally, the design would not detect absolute loosening. The paper focuses on Moody’s becoming more favorable relative to S&P.

### 4.3 Addressing issuer composition

- The model controls for issuer size, leverage, operating margin, stock volatility, issue size, maturity, and seniority.
- It allows control coefficients to differ before and after 2000.
- Robustness tests restrict the sample to firms issuing bonds in both periods.
- Additional tests include industry-year fixed effects and firm fixed effects.

**Interpretation:** These checks reduce concerns that the post-IPO result is driven only by changes in which firms issued bonds after 2000.

### 4.4 Addressing S&P conservatism

- The paper separately examines Moody’s ratings and S&P ratings as dependent variables.
- Moody’s ratings become more favorable after the IPO.
- S&P ratings do not show corresponding evidence of becoming unusually tougher.

**Interpretation:** The relative shift is mainly attributed to Moody’s becoming more favorable, not S&P becoming harsher.

### 4.5 Why conflict proxies matter

- The main result is strongest where Moody’s had stronger client-related incentives.
- Structured-product issuers and financial firms link the corporate-bond results to the structured-finance culture discussed in testimony.
- Large issuers and marginal issuers represent clients for whom a favorable rating may be especially valuable.

- High Fitch-share industries capture competitive pressure.

**Interpretation:** Cross-sectional concentration of the effect strengthens the interpretation that market-share and client-conflict incentives mattered after the IPO.

#### 4.6 Why informativeness tests matter

- A more favorable Moody's rating could be benign if it reflected better information about default or recovery.
- The paper tests this using EDFs, bond defaults, and recovery rates.
- The favorable post-IPO Moody's ratings are not associated with lower default risk.

**Interpretation:** These tests separate rating leniency from improved predictive accuracy.

#### 4.7 What the paper cannot fully prove

- The setting is a single-event study: Moody's IPO.
- Moody's and S&P could both have changed absolute standards in ways the relative design cannot detect.
- There may be Moody's-specific omitted shocks around 2000.
- The paper observes ratings and defaults, not internal rating committee deliberations.
- The CDO sample is shorter and more limited than the corporate-bond sample.

**Interpretation:** The paper's strongest claim is not that public ownership mechanically causes laxity everywhere. It shows that Moody's IPO is associated with a measurable, client-sensitive, and not-more-informative relative loosening of ratings.

## 5 Tables and figure

### 5.1 Table 1 + Table 2: rating scale and sample descriptives

Read:

- Table 1 maps Moody's and S&P letter ratings into a 1–21 numerical scale and shows the distribution of new bond ratings.
- The sample is concentrated in investment-grade ratings, with median numerical rating of 6 for both agencies.
- Table 2 shows 8,505 new bond issues with mean issue size of about \$141 million and average time to maturity of 7.75 years.
- The average Moody's numerical rating is 8.17 and the average S&P numerical rating is 8.06.

**Economic message:** The rating scale defines the dependent variable. The descriptive statistics show that the sample is large and covers economically important bond issues.

**Table 1**

Numerical coding of rating categories and frequencies of such categories for new bond issues.

This table provides summary information on Moody's and S&P's credit ratings on our sample of 8,505 new corporate bonds issued over the period 1995–2005 (excluding 2000). Panel A presents the frequency distributions of our bonds across different rating categories by Moody's and S&P, and the numerical coding of each rating category. Panel B presents the mean and median of the numerical ratings of our sample bonds assigned by Moody's and S&P.

Panel A: Frequency distribution

|                          | Numeric rating    | Moody's              |               | S&P                  |               |
|--------------------------|-------------------|----------------------|---------------|----------------------|---------------|
|                          |                   | Credit rating letter | Frequency (%) | Credit rating letter | Frequency (%) |
| Investment-grade         |                   |                      |               |                      |               |
| Highest quality          | 1                 | Aaa                  | 0.21          | AAA                  | 0.25          |
| Very high quality        | 2                 | Aa1                  | 0.07          | AA+                  | 0.68          |
|                          | 3                 | Aa2                  | 0.92          | AA                   | 1.61          |
|                          | 4                 | Aa3                  | 17.32         | AA−                  | 9.68          |
| High quality             | 5                 | A1                   | 11.13         | A+                   | 11.57         |
|                          | 6                 | A3                   | 22.54         | A                    | 32.58         |
|                          | 7                 | A3                   | 5.62          | A−                   | 4.83          |
| Minimum investment grade | 8                 | Baa1                 | 8.00          | BBB+                 | 5.02          |
|                          | 9                 | Baa2                 | 4.67          | BBB                  | 4.84          |
|                          | 10                | Baa3                 | 3.01          | BBB−                 | 3.14          |
|                          |                   |                      |               |                      |               |
| High-yield               |                   |                      |               |                      |               |
| Low grade                | 11                | Ba1                  | 1.72          | BB+                  | 1.63          |
|                          | 12                | Ba2                  | 1.27          | BB                   | 7.88          |
|                          | 13                | Ba3                  | 6.67          | BB−                  | 1.65          |
| Very speculative         | 14                | B1                   | 5.03          | B+                   | 3.64          |
|                          | 15                | B3                   | 4.96          | B                    | 5.08          |
|                          | 16                | B3                   | 5.2           | B−                   | 4.26          |
| Substantial risk         | 17                | Caa1                 | 1.00          | CCC+                 | 0.93          |
|                          | 18                | Caa2                 | 0.47          | CCC                  | 0.55          |
|                          | 19                | Caa3                 | 0.16          | CCC−                 | 0.12          |
|                          | Very poor quality | 20                   | Ca            | 0.04                 | CC            |
| 21                       |                   | C                    | 0.00          | C                    | 0.00          |
|                          |                   |                      |               |                      |               |

Panel B: Summary statistics on numerical ratings

|                  | Moody's |        | S&P   |        |
|------------------|---------|--------|-------|--------|
|                  | Mean    | Median | Mean  | Median |
| Full sample      | 8.17    | 6      | 8.06  | 6      |
| Investment-grade | 5.97    | 6      | 6.03  | 6      |
| High-yield       | 14.26   | 14     | 13.89 | 14     |

**Table 2**

Descriptive statistics of firms issuing rated new bonds.

The table presents summary information on our sample of 8,505 new corporate bonds issued over the period 1995–2005 (excluding 2000). Issuer size is the market value of equity plus the book value of debt. Leverage is long-term debt divided by total assets. Operating margin is operating income before depreciation divided by sales. Stock return standard deviation is the standard deviation of daily stock returns in the year prior. Issue size is the par value of the bond issue. Moody's and S&P's ratings are the numeric values of the ratings assigned by Moody's and S&P, coded as per Table 1. All firm characteristics are measured the year prior to the new issue.

|                                      | Mean      | Median    | Std.       |
|--------------------------------------|-----------|-----------|------------|
| Issuer size (\$ million)             | 93,589.51 | 38,808.46 | 117,998.88 |
| Leverage                             | 0.27      | 0.21      | 0.19       |
| Operating margin                     | 0.22      | 0.43      | 6.58       |
| Stock return standard deviation      | 0.06      | 0.02      | 0.15       |
| Issue size (\$ million)              | 141.40    | 50.00     | 291.00     |
| Time to maturity at issuance (years) | 7.75      | 5.76      | 8.03       |
| Moody's ratings                      | 8.17      | 6         | 4.04       |
| S&P ratings                          | 8.06      | 6         | 3.84       |

## 5.2 Table 3 + Figure 1: the basic pre/post shift

Read:

- Table 3 shows that before the IPO, Moody's is tougher than S&P: the full-sample mean *RatingDiff* is  $-0.302$ .
- After the IPO, Moody's is more favorable than S&P: the full-sample mean *RatingDiff* is 0.286.
- The pre/post difference is statistically significant at the 1% level.
- The shift appears for investment-grade and high-yield issues.
- Figure 1 shows the annual pattern: negative *RatingDiff* during 1995–1999 and positive *RatingDiff* after 2001, with stronger positive values in 2003–2005.

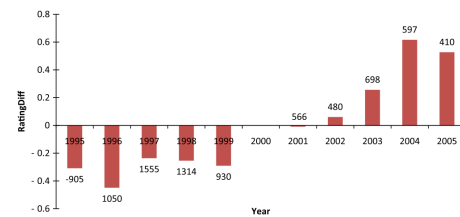
**Economic message:** This is the core visual and univariate evidence: Moody's relative stance changes sign after public listing.

**Table 3**

Univariate comparisons of ratings of new issues between Moody's and S&amp;P.

This table presents summary information on Moody's and S&P's respective numerical ratings of our sample of 8,505 new bond issues over the period 1994–2005 (excluding 2000), and the variable *RatingDiff*, which is S&P's numerical rating minus Moody's numerical rating for each bond, coded as per the legend to Table 1. As smaller numbers mean higher ratings, a positive value of *RatingDiff* implies that Moody's assigns a higher rating than S&P. The column "Pre-Moody's IPO" covers all eligible new bonds issued over 1995–1999. The column "Post-Moody's IPO" includes all eligible new bond issues over 2001–2005. The column "Difference test" presents the *p*-values related to the test of the difference in *RatingDiff* between the Pre- and the Post-Moody's IPO periods. "Investment-grade category (IV)" includes all new issues where both Moody's and S&P assigned an investment-grade rating at the time of issuance. "High-yield (HY)" refers to new issues where both Moody's and S&P assigned a high-yield rating at the time of issuance. "Across IV and HY" refers to the small sample of new issues where one rating agency assigns an investment-grade rating while the other assigns a high-yield rating.

|                       | Pre-Moody's IPO |              |                   | Post-Moody's IPO |              |                   | Difference test ( <i>p</i> -value) |
|-----------------------|-----------------|--------------|-------------------|------------------|--------------|-------------------|------------------------------------|
|                       | Moody's rating  | S&P's rating | <i>RatingDiff</i> | Moody's rating   | S&P's rating | <i>RatingDiff</i> |                                    |
| Full sample           |                 |              |                   |                  |              |                   |                                    |
| Mean                  | 8.515           | 8.213        | −0.302            | 7.455            | 7.741        | 0.286             | 0.000                              |
| Median                | 7.000           | 6.000        | 0.000             | 6.000            | 6.000        | 0.000             | 0.000                              |
| Nobs                  | 5,722           | 5,722        | 5,722             | 2,783            | 2,783        | 2,783             |                                    |
| Investment-grade (IV) |                 |              |                   |                  |              |                   |                                    |
| Mean                  | 6.235           | 6.046        | −0.189            | 5.423            | 5.847        | 0.425             | 0.000                              |
| Median                | 6.000           | 6.000        | 0.000             | 5.000            | 6.000        | 0.000             | 0.000                              |
| Nobs                  | 4,076           | 4,076        | 4,076             | 2,150            | 2,150        | 2,150             |                                    |
| High-yield (HY)       |                 |              |                   |                  |              |                   |                                    |
| Mean                  | 14.273          | 13.706       | −0.567            | 14.665           | 14.507       | −0.158            | 0.000                              |
| Median                | 14.000          | 14.000       | −1.000            | 15.000           | 15.000       | 0.000             | 0.000                              |
| Nobs                  | 1,588           | 1,588        | 1,588             | 582              | 582          | 582               |                                    |
| Across IV and HY      |                 |              |                   |                  |              |                   |                                    |
| Mean                  | 11.069          | 10.103       | −0.966            | 10.843           | 10.353       | −0.490            | 0.057                              |
| Median                | 11.000          | 10.000       | −1.000            | 11.000           | 10.000       | −1.000            | 0.037                              |
| Nobs                  | 58              | 58           | 58                | 51               | 51           | 51                |                                    |



RatingDiff = S&P's rating minus Moody's rating

## 5.3 Table 4 + Table 5: main new-issue regression and structured-product issuance

Read:

- Table 4 shows that the post-IPO dummy is positive and highly significant for new issues.
- Without controls, the post-IPO coefficient on *RatingDiff* is 0.585.

- With controls and interactions, the post-IPO coefficient rises to 0.851.
- When Moody's and S&P numerical ratings are separately used as outcomes, the post-IPO shift is mainly driven by more favorable Moody's ratings.
- Table 5 documents the rapid growth of structured-product issuance from \$142.65 billion in 1995 to \$1,605.60 billion in 2005.
- The top 40 structured-product issuers account for a large share of the market each year.

**Economic message:** The main regression confirms the univariate result after controls. Table 5 motivates why structured-product relationships could have been especially valuable to a public Moody's.

Moody's IPO. *RatingDiff* is the S&P's numerical rating minus the Moody's numerical rating for new bond issues, coded as per Table 1. As smaller numbers correspond to higher ratings, a positive (negative) value of *RatingDiff* implies that Moody's assigns a higher (lower) rating than S&P. The number on top of each bar represents the number of new bond issues over which the *RatingDiff* variable was computed every year.

**Table 4**

Did Moody's assign higher ratings for new issues after its IPO?

The sample consists of 8,505 new bonds issued between 1995 and 2005 excluding 2000. The dependent variable in columns 1–2 is *RatingDiff*. In columns 3 and 4, it is the numerical rating by Moody's and S&P, respectively. *RatingDiff* is S&P rating minus Moody's rating. The *post2000Dum* is one for bonds issued during the post-public period and zero otherwise. *IssuerSize* is the natural log of market value. *Leverage* is ratio of long-term debt to total assets. *OpMargin* is operating income before depreciation divided by sales. *Stkretstd* is the standard deviation of daily stock returns in the year prior to the issuance. *IssueSize* is the logarithm of the par value of the bond issue. Variables are measured in the year prior to the new issue. Heteroskedasticity-adjusted robust *p*-values are in parentheses. We cluster standard errors by the issuing firm. The superscript \*\*\*, \*\*, \* represents significance at 1%, 5%, and 10% level, respectively.

|                                        | (1)<br><i>RatingDiff</i> | (2)<br><i>RatingDiff</i> | (3)<br>Moody's rating | (4)<br>S&P's rating   |
|----------------------------------------|--------------------------|--------------------------|-----------------------|-----------------------|
| Intercept                              | –0.299<br>(0.000)***     | 0.155<br>(0.531)         | 8.401<br>(0.000)***   | 8.556<br>(0.000)***   |
| <i>post2000Dum</i>                     | 0.585<br>(0.000)***      | 0.851<br>(0.000)***      | –0.935<br>(0.000)***  | –0.084<br>(0.888)     |
| <i>IssuerSize</i>                      | –0.010<br>(0.862)        | –0.010<br>(0.862)        | –1.259<br>(0.000)***  | –1.269<br>(0.000)***  |
| <i>Leverage</i>                        | –0.105<br>(0.719)        | –0.105<br>(0.719)        | 2.755<br>(0.010)***   | 2.651<br>(0.015)**    |
| <i>OpMargin</i>                        | –0.006<br>(0.000)***     | –0.006<br>(0.000)***     | –0.026<br>(0.000)***  | –0.032<br>(0.000)***  |
| <i>Stkretstd</i>                       | –1.314<br>(0.000)***     | –1.314<br>(0.000)***     | 7.397<br>(0.000)***   | 6.083<br>(0.000)***   |
| <i>IssueSize</i>                       | 0.034<br>(0.290)         | 0.034<br>(0.290)         | 0.193<br>(0.119)      | 0.227<br>(0.031)**    |
| <i>YTM</i>                             | –16.560<br>(0.220)       | –16.560<br>(0.220)       | –65.638<br>(0.112)    | –82.198<br>(0.033)**  |
| <i>SeniorDum</i>                       | –0.188<br>(0.072)*       | –0.188<br>(0.072)*       | –1.673<br>(0.000)***  | –1.860<br>(0.000)***  |
| <i>IssuerSize</i> × <i>post2000Dum</i> | 0.123<br>(0.043)**       | 0.123<br>(0.043)**       | 0.041<br>(0.747)      | 0.164<br>(0.151)      |
| <i>Leverage</i> × <i>post2000Dum</i>   | –0.052<br>(0.881)        | –0.052<br>(0.881)        | 1.290<br>(0.347)      | 1.238<br>(0.372)      |
| <i>OpMargin</i> × <i>post2000Dum</i>   | 0.006<br>(0.000)***      | 0.006<br>(0.000)***      | 0.015<br>(0.000)***   | 0.021<br>(0.000)***   |
| <i>Stkretstd</i> × <i>post2000Dum</i>  | 1.725<br>(0.641)         | 1.725<br>(0.641)         | 9.575<br>(0.356)      | 11.300<br>(0.313)     |
| <i>IssueSize</i> × <i>post2000Dum</i>  | –0.106<br>(0.012)**      | –0.106<br>(0.012)**      | –0.012<br>(0.920)     | –0.118<br>(0.229)     |
| <i>YTM</i> × <i>post2000Dum</i>        | 6.859<br>(0.759)         | 6.859<br>(0.759)         | 124.945<br>(0.002)*** | 131.804<br>(0.001)*** |
| <i>SeniorDum</i> × <i>post2000Dum</i>  | 0.019<br>(0.897)         | 0.019<br>(0.897)         | –0.096<br>(0.830)     | –0.077<br>(0.850)     |
| Adjusted R-square                      | 0.095                    | 0.216                    | 0.765                 | 0.762                 |
| ..                                     | 0.000                    | 0.000                    | 0.000                 | 0.000                 |

**Table 5**

Summary statistics on structured products issuance.

This table provides summary information on the issuance of structured finance products from 1995–2005. The structured products included are asset-backed securities (ABS), mortgage-backed securities (MBS), and collateralized debt obligations (CDOs). The data come from the ABS database managed by J.P. Morgan's Asset Backed Alert.

| Year | Total issuance<br>(\$billion) | Number of<br>issues | Number of<br>issuers | Share of top 40<br>issuers |
|------|-------------------------------|---------------------|----------------------|----------------------------|
| 1995 | 142.65                        | 492                 | 208                  | 0.75                       |
| 1996 | 212.01                        | 684                 | 279                  | 0.68                       |
| 1997 | 322.49                        | 903                 | 328                  | 0.62                       |
| 1998 | 438.46                        | 1015                | 352                  | 0.61                       |
| 1999 | 416.80                        | 1002                | 358                  | 0.57                       |
| 2000 | 399.63                        | 938                 | 315                  | 0.59                       |
| 2001 | 529.90                        | 1157                | 308                  | 0.69                       |
| 2002 | 688.39                        | 1521                | 286                  | 0.76                       |
| 2003 | 861.18                        | 1760                | 253                  | 0.78                       |
| 2004 | 1115.81                       | 1948                | 266                  | 0.80                       |

## 5.4 Table 6 + Table 7: cross-sectional conflicts and outstanding bonds

Read:

- Table 6 shows post-IPO relative Moody's generosity is stronger for high-conflict new issues.
- The interaction terms are positive for structured-product issuers, financial firms, large/frequent bond issuers, marginal issuers, and high-Fitch-share industries.
- Table 7 repeats the analysis for outstanding bonds using *LeadTimeDiff*.
- The post-IPO dummy is positive, and several high-conflict interactions remain positive and significant.

**Economic message:** The effect is not uniform. It is strongest where market-share and client relationship incentives are most salient.

## 5.5 Table 8 + Table 9 + Table 10: informativeness and alternative explanations

Read:

- Table 8 tests whether Moody's higher post-IPO relative ratings are associated with lower default risk.

**Table 6**

Cross-sectional variation in Moody's ratings post-IPO for new issues.

This table displays partial results for the ordinary least squares (OLS) estimation where the dependent variable was *RatingDiff*, the S&P numerical rating minus the Moody's numerical rating. The sample consists of 8,505 new bonds issued between 1995 and 2005 excluding 2000. The *post2000Dum* takes the value of one if the bond is issued between 2001 and 2005 and zero otherwise. In column 1 (2), *HighConfDum* takes the value one if the bond is issued by firms that are large issuers of structured finance products (financial firms). In column 3 (4) *HighConfDum* takes the value one if the bond is issued by a large issuer (the issuer's profit margin is above the median for that S&P rating grade). In column 5, *HighConfDum* takes the value one if Fitch's market share in the two-digit FSD industry is above the median value for the year. Other variables included but not reported in the table are *IssuerSize* (natural log of total market value), *Leverage* (ratio of long-term debt to total assets), *OpMargin* (operating income before depreciation divided by sales), *Stkretstd* (standard deviation of daily stock returns in the year prior to the issuance), *IssuerLog* (log of the par value of the bond issue), *YTM* (number of years to maturity), *SeniorDum* (dummy variable that is one for senior debt). All accounting variables are measured in the year prior to the new issue. Heteroskedasticity-adjusted robust *p*-values are in parentheses. We cluster standard errors by the issuing firm. The superscript \*\*\*, \*\*, \* represents significance at 1%, 5%, and 10% level, respectively.

|                                         | (1)                                                                     | (2)                 | (3)                  | (4)                      | (5)                     |
|-----------------------------------------|-------------------------------------------------------------------------|---------------------|----------------------|--------------------------|-------------------------|
| Issuer characteristics                  | Structured products                                                     | Finance firms       | Large and frequent   | Missed S&P higher rating | High Fitch market share |
| Intercept                               | 0.133<br>(0.193)                                                        | 0.047<br>(0.684)    | 0.291<br>(0.002)***  | 0.228<br>(0.009)***      | 0.277<br>(0.006)***     |
| <i>post2000Dum</i>                      | 0.632<br>(0.000)***                                                     | 1.036<br>(0.000)*** | 0.699<br>(0.000)***  | 0.744<br>(0.000)***      | 0.246<br>(0.019)**      |
| <i>HighConfDum</i>                      | 0.024<br>(0.867)                                                        | 0.320<br>(0.000)*** | -0.205<br>(0.000)*** | -0.415<br>(0.003)***     | -0.122<br>(0.000)***    |
| <i>HighConfDum</i> × <i>post2000Dum</i> | 0.212<br>(0.000)***                                                     | 0.320<br>(0.000)*** | 0.308<br>(0.000)***  | 0.368<br>(0.000)***      | 0.567<br>(0.000)***     |
| Adjusted R-square                       | Control variables have been included in the estimation but not reported |                     |                      |                          |                         |
| <i>N</i>                                | 8,505                                                                   | 8,505               | 8,505                | 8,505                    | 8,505                   |

**Table 7**

Moody's higher ratings for all bond issues after its IPO.

This table presents results from the OLS regressions where the dependent variable is *LeadTimeDiff*, the fraction of a year where Moody's assigns a higher rating minus the fraction where S&P assigns a higher rating. The sample consists of 81,641 bond-years for outstanding bonds over the period 1995–2005 excluding 2000. The *post2000Dum* is a dummy variable that takes the value of one for the years in the post-public period, and zero otherwise. *HighConfDum* takes the value one if issuer is a large issuer of structured products (column 2), a finance firm (column 3), a large issuer of corporate bonds (column 4), missed a higher S&P rating (column 5), and in industry with above median Fitch market share (column 6). Other variables are defined as in Table 4. All models include a constant which has not been reported for brevity. Heteroskedasticity-adjusted robust *p*-values are in parentheses. We cluster standard errors by the issuing firm. The superscript \*\*\*, \*\*, \* represents significance at 1%, 5%, and 10% level, respectively.

|                                         | (1)                  | (2)                  | (3)                  | (4)                  | (5)                      | (6)                     |
|-----------------------------------------|----------------------|----------------------|----------------------|----------------------|--------------------------|-------------------------|
|                                         | Base                 | Structured products  | Finance issuers      | Large issuers        | Missed S&P higher rating | High Fitch market share |
| <i>post2000Dum</i>                      | 0.301<br>(0.000)***  | 0.187<br>(0.000)***  | 0.165<br>(0.000)***  | 0.257<br>(0.000)***  | 0.229<br>(0.000)***      | 0.053<br>(0.000)***     |
| <i>HighConfDum</i>                      |                      | 0.026<br>(0.011)**   | 0.227<br>(0.000)***  | -0.128<br>(0.000)*** | 0.011<br>(0.085)*        | -0.097<br>(0.000)***    |
| <i>HighConfDum</i> × <i>post2000Dum</i> |                      | 0.323<br>(0.000)***  | 0.445<br>(0.000)***  | 0.052<br>(0.010)**   | 0.152<br>(0.000)***      | 0.265<br>(0.000)***     |
| <i>IssuerSize</i>                       | 0.016<br>(0.000)***  | 0.013<br>(0.000)***  | 0.001<br>(0.693)     | 0.020<br>(0.000)***  | 0.015<br>(0.000)***      | 0.025<br>(0.000)***     |
| <i>Leverage</i>                         | 0.087<br>(0.000)***  | 0.083<br>(0.000)***  | 0.046<br>(0.034)**   | 0.114<br>(0.000)***  | 0.083<br>(0.000)***      | 0.102<br>(0.000)***     |
| <i>OpMargin</i>                         | -0.005<br>(0.000)*** | -0.005<br>(0.000)*** | -0.005<br>(0.000)*** | -0.005<br>(0.000)*** | -0.005<br>(0.000)***     | -0.005<br>(0.000)***    |
| <i>Stkretstd</i>                        | -0.246<br>(0.011)**  | -0.264<br>(0.005)*** | -0.159<br>(0.118)    | -0.255<br>(0.008)*** | -0.244<br>(0.011)**      | -0.295<br>(0.002)**     |
| <i>IssuerSize</i>                       | 0.005<br>(0.001)***  | 0.005<br>(0.001)***  | 0.005<br>(0.001)***  | 0.005<br>(0.001)***  | 0.005<br>(0.001)***      | 0.004<br>(0.004)***     |
| <i>YTM</i>                              | 0.019<br>(0.000)***  | 0.020<br>(0.000)***  | 0.026<br>(0.000)***  | 0.019<br>(0.000)***  | 0.019<br>(0.000)***      | 0.017<br>(0.000)***     |
| <i>SeniorDum</i>                        | -0.044<br>(0.000)*** | -0.043<br>(0.000)*** | 0.004<br>(0.696)     | -0.036<br>(0.000)*** | -0.044<br>(0.000)***     | -0.045<br>(0.000)***    |
| <i>IssuerSize</i> × <i>post2000Dum</i>  | 0.099<br>(0.000)***  | 0.070<br>(0.000)***  | 0.085<br>(0.000)***  | 0.098<br>(0.000)***  | 0.093<br>(0.000)***      | 0.073<br>(0.000)***     |
| <i>Leverage</i> × <i>post2000Dum</i>    | -0.538<br>(0.000)*** | -0.488<br>(0.000)*** | -0.418<br>(0.000)*** | -0.554<br>(0.000)*** | -0.543<br>(0.000)***     | -0.631<br>(0.000)***    |
| <i>OpMargin</i> × <i>post2000Dum</i>    | 0.006<br>(0.000)***  | 0.006<br>(0.000)***  | 0.005<br>(0.000)***  | 0.006<br>(0.000)***  | 0.006<br>(0.000)***      | 0.006<br>(0.000)***     |
| <i>Stkretstd</i> × <i>post2000Dum</i>   | 0.225<br>(0.020)**   | 0.241<br>(0.011)**   | 0.144<br>(0.158)     | 0.234<br>(0.016)**   | 0.226<br>(0.020)**       | 0.277<br>(0.004)***     |
| <i>IssuerSize</i> × <i>post2000Dum</i>  | 0.009<br>(0.000)***  | 0.009<br>(0.000)***  | 0.021<br>(0.000)***  | 0.009<br>(0.000)***  | 0.006<br>(0.007)***      | 0.017<br>(0.000)***     |
| <i>YTM</i> × <i>post2000Dum</i>         | -0.005<br>(0.300)    | 0.009<br>(0.060)*    | -0.029<br>(0.000)*** | -0.004<br>(0.411)    | -0.010<br>(0.338)**      | 0.003<br>(0.457)        |
| <i>SeniorDum</i> × <i>post2000Dum</i>   | -0.158<br>(0.000)*** | -0.127<br>(0.000)*** | 0.004<br>(0.761)     | -0.134<br>(0.000)*** | -0.117<br>(0.000)***     | -0.12<br>(0.000)***     |
| Adjusted R-square                       | 0.093                | 0.106                | 0.125                | 0.095                | 0.187                    | 0.104                   |
| <i>N</i>                                | 81,641               | 81,641               | 81,641               | 81,641               | 81,641                   | 81,641                  |

- EDF and default interactions do not show that post-IPO Moody's relative ratings became more informative.
- Table 9 shows no meaningful pre/post difference in recovery rates around default.
- Table 10 shows that the post-IPO result survives in industries with little change in Fitch competition.

**Economic message:** The post-IPO rating tilt does not appear to be justified by superior default prediction, higher recovery, or Fitch's expansion alone.

**Table 8**

Alternate measure of rating accuracy.

The New issues sample consists of 8,505 new bond, and the All issues consists of 81,641 bond-years, both over the period 1995–2005 excluding 2000. The dependent variable for the New issues sample is *RatingDiff*, and it is *LeadTimeDiff* for the All issues sample. The *post2000Dum* is a dummy that takes the value one for years after 2000, and zero otherwise. The first two columns use *EDF* which is the expected default frequency estimated using the Black-Scholes-Merton specification for an issuing firm in the sample. The third column uses *DefaultDum*, a dummy that takes the value one if the bond defaults in the next two years. The remaining variables are as defined in Table 4. The number below each estimate of the coefficients is heteroskedasticity-adjusted robust *p*-value. We cluster standard errors by the issuing firm. The superscript \*\*\*, \*\*, \* represents significance at 1%, 5%, and 10% level, respectively.

|                                                       | (1) New issues<br>EDF | (2) All issues<br>EDF | (3) All issues<br>Bond default |
|-------------------------------------------------------|-----------------------|-----------------------|--------------------------------|
| Intercept                                             | 0.251<br>(0.260)      | -0.022<br>(0.009)***  | -0.041<br>(0.000)***           |
| <i>post2000Dum</i>                                    | 0.623<br>(0.000)***   | 0.301<br>(0.000)***   | 0.298<br>(0.000)***            |
| <i>EDF</i> ( <i>DefaultDum</i> )                      | -0.404<br>(0.000)***  | -0.153<br>(0.000)***  | 0.020<br>(0.530)               |
| <i>EDF</i> ( <i>DefaultDum</i> ) × <i>post2000Dum</i> | 0.366<br>(0.099)*     | 0.064<br>(0.013)***   | 0.176<br>(0.000)***            |
| <i>IssuerSize</i>                                     | 0.005<br>(0.702)      | 0.017<br>(0.000)***   | 0.016<br>(0.000)***            |
| <i>Leverage</i>                                       | 0.040<br>(0.740)      | 0.105<br>(0.000)***   | 0.085<br>(0.000)***            |
| <i>OpMargin</i>                                       | -0.005<br>(0.000)***  | -0.063<br>(0.628)     | -0.005<br>( $< .0001$ )***     |
| <i>Stkretstd</i>                                      | -0.355<br>(0.123)     | -0.005<br>(0.000)***  | -0.249<br>(0.010)**            |
| <i>IssueSize</i>                                      | 0.057<br>(0.001)***   | 0.006<br>(0.000)***   | 0.005<br>(0.001)***            |
| <i>YTM</i>                                            | -18.612<br>(0.072)*   | 0.014<br>(0.000)***   | 0.019<br>(0.000)***            |
| <i>SeniorDum</i>                                      | -0.158<br>(0.001)***  | -0.044<br>(0.000)***  | -0.044<br>(0.000)***           |
| <i>IssuerSize</i> × <i>post2000Dum</i>                | 0.079<br>(0.000)***   | 0.098<br>(0.000)***   | 0.099<br>(0.000)***            |
| <i>Leverage</i> × <i>post2000Dum</i>                  | -0.135<br>(0.488)     | -0.522<br>(0.000)***  | -0.552<br>(0.000)***           |
| <i>OpMargin</i> × <i>post2000Dum</i>                  | 0.006<br>(0.000)***   | 0.054<br>(0.680)      | 0.006<br>(0.000)***            |
| <i>Stkretstd</i> × <i>post2000Dum</i>                 | 3.317<br>(0.089)*     | 0.006<br>(0.000)***   | 0.228<br>(0.018)**             |
| <i>IssueSize</i> × <i>post2000Dum</i>                 | -0.159<br>(0.000)***  | 0.008<br>(0.000)***   | 0.009<br>(0.000)***            |
| <i>YTM</i> × <i>post2000Dum</i>                       | -12.112<br>(0.533)    | 0.000<br>(0.923)      | -0.005<br>(0.296)              |
| <i>SeniorDum</i> × <i>post2000Dum</i>                 | 0.090<br>(0.282)      | -0.129<br>(0.000)***  | -0.126<br>(0.000)***           |
| Adjusted R-square                                     | 0.205                 | 0.095                 | 0.094                          |
| <i>N</i>                                              | 8,505                 | 81,641                | 81,641                         |

**Table 9**

Recovery rates.

This table displays recovery rates for bonds in default before and after Moody's IPO in 2000. In Panel A (B) recovery rate is estimated as the ratio of the market bid price around 30 (45) days after default to the par value. We test the difference in recovery rates in the Pre-Moody's IPO period (1995–1999) from that in the Post-Moody's IPO period (2001–2005) and report the  $p$ -values in the column “Difference test.”

|                                               | Pre-Moody's IPO | Post-Moody's IPO | Difference test<br>( $p$ -value) |
|-----------------------------------------------|-----------------|------------------|----------------------------------|
| <i>Panel A: 30 Days following default (%)</i> |                 |                  |                                  |
| Mean                                          | 47.746          | 48.956           | 0.871                            |
| Median                                        | 44.500          | 44.210           | 0.933                            |
| Nobs                                          | 15              | 88               |                                  |
| <i>Panel B: 45 Days following default (%)</i> |                 |                  |                                  |
| Mean                                          | 40.204          | 46.887           | 0.377                            |
| Median                                        | 42.000          | 43.000           | 0.403                            |
| Nobs                                          | 15              | 88               |                                  |

## 5.6 Table 11 + Table 12: structured products and CDO ratings

Read:

- Table 11 describes the CDO sample and shows that most deals have high agreement on AAA tranche treatment, but disagreement is economically meaningful.
- In deals with all tranches rated by both agencies, 88% show agreement on AAA treatment and 6% show disagreement.
- Table 12 shows Moody's post-IPO CDO ratings are more favorable relative to S&P.
- In the restricted sample of deals with all tranches rated by both agencies, the deal-level difference rises from  $-0.023$  pre-IPO to  $0.026$  post-IPO.

**Economic message:** The pattern is not confined to corporate bonds. It also appears in structured finance, the area most associated with the congressional allegations.

**Table 11**

Summary information for CDOs.

Panel A displays summary statistics of the width of the first three tranches for CDOs issued between 1997 and 2003 (excluding 2000). Summary statistics are presented both for the sample of 308 deals with all tranches rated by both agencies, and the sample of 703 deals with at least one tranche rated by both agencies. Information for higher level tranches has not been reported for brevity. Panel B analyzes AAA tranche ratings disagreement between Moody's and S&P.

| <i>Panel A: Summary statistics of tranche width</i>                          |                                                |         |                                                        |         |                                                        |        |     |     |
|------------------------------------------------------------------------------|------------------------------------------------|---------|--------------------------------------------------------|---------|--------------------------------------------------------|--------|-----|-----|
| Tranche                                                                      | Deals with all tranches rated by both agencies |         |                                                        |         | Deals with at least one tranche rated by both agencies |        |     |     |
|                                                                              | Mean                                           | Median  | Std                                                    | N       | Mean                                                   | Median | Std | N   |
| 1                                                                            | 70%                                            | 77%     | 27%                                                    | 308     | 62%                                                    | 69%    | 27% | 703 |
| 2                                                                            | 19%                                            | 13%     | 16%                                                    | 233     | 18%                                                    | 12%    | 16% | 626 |
| 3                                                                            | 11%                                            | 7%      | 11%                                                    | 212     | 10%                                                    | 7%     | 10% | 589 |
| <i>Panel B: AAA tranche ratings disagreement between Moody's and S&amp;P</i> |                                                |         |                                                        |         |                                                        |        |     |     |
|                                                                              | Deals with all tranches rated by both agencies |         | Deals with at least one tranche rated by both agencies |         |                                                        |        |     |     |
|                                                                              | Number of deals                                | % Total | Number of deals                                        | % Total |                                                        |        |     |     |
| No AAA tranche                                                               | 17                                             | 6%      | 36                                                     | 5%      |                                                        |        |     |     |
| Agreement on AAA                                                             | 272                                            | 88%     | 586                                                    | 83%     |                                                        |        |     |     |
| Disagreement on AAA                                                          | 19                                             | 6%      | 81                                                     | 11%     |                                                        |        |     |     |
| Disagreement on 1st tranche                                                  | 12                                             | 63%     | 24                                                     | 30%     |                                                        |        |     |     |
| Disagreement on 2nd tranche                                                  | 5                                              | 23%     | 23                                                     | 29%     |                                                        |        |     |     |

**Table 10**

Impact of Fitch's expansion.

This table displays the estimation of the base model for new issues (column 1) and all issues (column 2) in a sample of industries with the least growth in Fitch competition from the pre-public period (1995–1999) to the post-public period (2001–2005). Competition from Fitch is captured as the market share of Fitch in all corporate bond ratings in the FSD two-digit industry. The least growth industries are those with below median growth rate in Fitch's market share from the pre- to the post-public period. Other variables included but not reported in the table are *IssuerSize* (natural log of total market value), *Leverage* (ratio of long-term debt to total assets), *OpMargin* (operating income before depreciation divided by sales), *Stkretstd* (standard deviation of daily stock returns in the year prior to the issuance), *IssueSize* (log of the par value of the bond issue), *YTM* (number of years to maturity), and *SeniorDum* (dummy variable that is one for senior debt). The interactions of all variables with the *post2000Dum* were also included. All accounting variables are measured in the year prior to the new issue. Heteroskedasticity-adjusted robust  $p$ -values are in parentheses. We cluster standard errors by the issuing firm. The superscript \*\*\*, \*\*, \* represents significance at 1%, 5%, and 10% level, respectively.

|                                              | (1) New issues      | (2) All issues      |
|----------------------------------------------|---------------------|---------------------|
| Intercept                                    | 0.173<br>(0.424)    | 0.082<br>(0.000)*** |
| <i>post2000Dum</i>                           | 0.819<br>(0.000)*** | 0.310<br>(0.000)*** |
| Control variables included but not tabulated |                     |                     |
| Adjusted R-square                            | 0.056               | 0.091               |

**Table 12**

Initial ratings on CDOs.

The sample consists of CDOs issued between 1997 and 2003 excluding 2000. Pre-Moody's IPO consists of CDOs issued over the period 1997–1999 and Post-Moody's IPO consists of deals issued over the period 2001–2003. Panel A includes deals where all tranches were rated by both Moody's and S&P. Panel B consists of deals with at least one tranche that is rated by both Moody's and S&P. The table displays the mean and median values of *RatingDiff*. For the deal-level analysis, *RatingDiff* is the difference in the fraction of the deal that is rated AAA by Moody's minus the fraction of the deal that is rated AAA by S&P. Fraction of the deal that is AAA is the number of tranches that are AAA-rated over the total number of tranches. For the tranche-level analysis, *RatingDiff* is the dummy that is one when Moody's rates the tranche as AAA minus the dummy when S&P rates the tranche as AAA. Nobs is the number of deals/tranches. Total issuance is the par value of the deals/tranches. The  $p$ -value is for test on the difference in Pre-Moody IPO and Post-Moody IPO mean and median. The superscript \*\*\*, \*\*, \* represents significance at 1%, 5%, and 10% level, respectively.

|                                                                                   | Total issuance (\$ billion) | Pre-Moody's IPO<br>1,243 | Post-Moody's IPO<br>2,309 | Difference test (p-value) |
|-----------------------------------------------------------------------------------|-----------------------------|--------------------------|---------------------------|---------------------------|
| <b>Panel A: Deal with all tranches rated by both Moody's and S&amp;P</b>          |                             |                          |                           |                           |
| Deal level                                                                        |                             |                          |                           |                           |
| Mean                                                                              | -0.023                      | 0.026                    | 0.008***                  |                           |
| Median                                                                            | 0                           | 0                        | 0.034**                   |                           |
| Nobs                                                                              | 39                          | 269                      |                           |                           |
| Total issuance                                                                    | 272                         | 1,131                    |                           |                           |
| Tranche level                                                                     |                             |                          |                           |                           |
| Mean                                                                              | -0.021                      | 0.007                    | 0.015**                   |                           |
| Median                                                                            | 0                           | 0                        | 0.008***                  |                           |
| Nobs                                                                              | 145                         | 1068                     |                           |                           |
| Total issuance                                                                    | 272                         | 1,131                    |                           |                           |
| <b>Panel B: Deals with at least one tranche rated by both Moody's and S&amp;P</b> |                             |                          |                           |                           |
| Deal level                                                                        |                             |                          |                           |                           |
| Mean                                                                              | -0.011                      | 0.010                    | 0.036**                   |                           |
| Median                                                                            | 0                           | 0                        | 0.124                     |                           |
| Nobs                                                                              | 158                         | 545                      |                           |                           |
| Total issuance                                                                    | 970                         | 1,987                    |                           |                           |
| Tranche level                                                                     |                             |                          |                           |                           |
| Mean                                                                              | -0.016                      | -0.002                   | 0.042**                   |                           |
| Median                                                                            | 0                           | 0                        | 0.020**                   |                           |

## 5.7 Table 13 + Table 14: robustness and cross-sectional tests

Read:

- Table 13 shows the result survives multiple robustness checks: longer windows, shorter windows, Fitch as benchmark, same issuing firms, industry-year fixed effects, firm fixed effects, and one bond per firm.
- The post-IPO coefficient remains positive across specifications.
- Table 14 estimates cross-sectional intercepts by year or quarter.

- Post-public intercepts are significantly higher than pre-public intercepts for new issues and all outstanding issues.

**Economic message:** The robustness evidence reduces concerns that the result is driven by the crisis, window choice, issuer composition, serial correlation, or benchmarking only to S&P.

**Table 13**

Robustness tests.

The sample period for Panel A is 1991–2009–2009, while for Panel B is 1999–2001–2001. Panel C presents the results when Fitch's ratings are used as the benchmark over the period 1995–2005. Panel D restricts the sample to bonds by issuing firms that issued (had outstanding) bonds in both the pre- and post-public periods. Panel E includes industry-year fixed effects. Panel F includes firm fixed effects when the firm issues two or more bond issues. Panel G focuses on a sample of firms with one bond in each period. We estimate Model (1) on the sample of new bond issues, and Model (2) on the sample that includes all outstanding issues. The dependent variables for Model (1) and Model (2) are *RatingDiff* and *LeadTimeDiff*, respectively. *RatingDiff* is the S&P numerical rating minus the Moody's numerical rating. *LeadTimeDiff* is the fraction of a year where Moody's assigns a higher rating minus the fraction of a year where S&P assigns a higher rating. The *post2000Dum* is a dummy variable that takes the value of one for the years in the post-public period, and zero otherwise. Other variables included in the estimation but not displayed are *IssuerSize*, *Leverage*, *OpMargin*, *Sktrestsd*, *IssueSize*, *YTM*, and *SeniorDum*. The interactions of all variables with the *post2000Dum* were also included. All accounting variables are measured in the year prior to the new issue. Heteroskedasticity-adjusted robust *p*-values are in parentheses. We cluster standard errors by the issuing firm. \*\*\*, \*\*, \* Represent significance at 1%, 5%, and 10% level, respectively.

|                    | Panel A: 9 Years around 2000 |                      | Panel B: 1 Year around 2000                  |                      | Panel C: Using Fitch as the benchmark |                      |
|--------------------|------------------------------|----------------------|----------------------------------------------|----------------------|---------------------------------------|----------------------|
|                    | (1) New issues               | (2) All issues       | (1) New issues                               | (2) All issues       | (1) New issues                        | (2) All issues       |
| Intercept          | −0.251<br>(0.002)***         | −0.051<br>(0.000)*** | −0.023<br>(0.931)                            | −0.162<br>(0.000)*** | −1.724<br>(0.000)***                  | −0.828<br>(0.000)*** |
| <i>post2000Dum</i> | 0.178<br>(0.000)***          | 0.285<br>(0.000)***  | 0.475<br>(0.049)**                           | 0.114<br>(0.000)***  | 0.555<br>(0.023)**                    | 0.106<br>(0.000)***  |
| Adjusted R-square  | 0.145                        | 0.090                | Control variables included but not tabulated |                      |                                       |                      |
| N                  | 11,697                       | 137,411              | 0.092<br>1,496                               | 0.042<br>15,827      | 0.194<br>5,851                        | 0.117<br>32,428      |
|                    | Panel D: Same issuing firms  |                      | Panel E: Industry-year fixed effects         |                      | Panel F: Firm fixed effects           |                      |
|                    | (1) New issues               | (2) All issues       | (1) New issues                               | (2) All issues       | (1) New issues                        | (2) All issues       |
| Intercept          | 1.098<br>(0.035)**           | −0.074<br>(0.000)*** | 0.096<br>(0.855)                             | −0.079<br>(0.329)    | 0.069<br>(0.858)                      | −0.335<br>(0.000)*** |
| <i>post2000Dum</i> | 1.028<br>(0.001)***          | 0.437<br>(0.000)***  | 0.466<br>(0.096)*                            | 0.221<br>(0.010)***  | 1.105<br>(0.008)***                   | 0.095<br>(0.000)***  |
| Adjusted R-square  | 0.316                        | 0.100                | Control variables included but not tabulated |                      |                                       |                      |
| N                  | 4,705                        | 57,124               | 0.343<br>8,505                               | 0.216<br>81,641      | 0.642<br>8,505                        | 0.489<br>81,641      |
|                    | Panel G: One bond per firm   |                      |                                              |                      |                                       |                      |
|                    | (1) New issues               | (2) All issues       |                                              |                      |                                       |                      |
| Intercept          | −2.618<br>(0.064)*           | 0.000<br>(0.999)     |                                              |                      |                                       |                      |
| <i>post2000Dum</i> | 3.062<br>(0.044)**           | 0.260<br>(0.079)*    |                                              |                      |                                       |                      |
| Adjusted R-square  | 0.001                        | 0.018                | Control variables included but not tabulated |                      |                                       |                      |
| N                  | 296                          | 3,586                |                                              |                      |                                       |                      |

To address this issue, we re-estimate our results using a longer short time period, i.e., from 1999–2001, considering rat-

**Table 14**

Cross-sectional tests.

This table presents summary statistics for the intercept term estimated in cross-sectional regressions over the sample period 1995–2005, not including 2000. The New issues sample consists of 8,505 new bond, and the All issues consists of 81,641 bond-years within our sample period. Panel A (B) reports summary for model estimated at the annual (quarterly) level. The dependent variable is *RatingDiff* for the new issues sample and *LeadTimeDiff* for the all issues sample. We calculate the mean of the estimates of the intercept for the pre-public and the post-public periods separately. We conduct tests for the difference in mean and median values from these two periods and report the *p*-values in the table. \*\*\*, \*\*, \* Represent significance at 1%, 5%, and 10% level, respectively.

|                              | Panel A: Annually |            | Panel B: Quarterly |            |
|------------------------------|-------------------|------------|--------------------|------------|
|                              | New issues        | All issues | New issues         | All issues |
| Pre-public period            |                   |            |                    |            |
| Mean                         | 0.170             | −0.046     | 0.412              | −0.022     |
| Median                       | 0.052             | −0.014     | 0.313              | −0.034     |
| Std                          | 0.372             | 0.073      | 0.791              | 0.044      |
| N                            | 5                 | 5          | 20                 | 20         |
| Post-public period           |                   |            |                    |            |
| Mean                         | 1.416             | 0.286      | 1.395              | 0.163      |
| Median                       | 1.553             | 0.356      | 1.472              | 0.189      |
| Std                          | 1.104             | 0.232      | 1.405              | 0.234      |
| N                            | 5                 | 5          | 20                 | 20         |
| Test on difference in mean   |                   |            |                    |            |
| <i>p</i> -value              | 0.063*            | 0.030**    | 0.032**            | 0.002***   |
| Test on difference in median |                   |            |                    |            |
| <i>p</i> -value              | 0.060*            | 0.060*     | 0.039**            | 0.004***   |

## 6 Short summary

- The paper asks whether Moody's public listing impaired the quality of its credit ratings.
- The main research design compares Moody's ratings to S&P ratings before and after Moody's 2000 IPO.
- The corporate bond sample contains 8,505 new bond issues rated by both Moody's and S&P during 1995–2005, excluding 2000.
- *RatingDiff* is defined so that positive values mean Moody's rates the issue more favorably than S&P.
- Before the IPO, the mean *RatingDiff* is −0.302, meaning Moody's is tougher than S&P.
- After the IPO, the mean *RatingDiff* is 0.286, meaning Moody's is more favorable than S&P.
- The shift of about 0.588 notches is statistically and economically significant.
- Regression controls do not eliminate the shift; the controlled post-IPO coefficient is even larger.
- The shift appears for both investment-grade and high-yield bonds.
- The evidence points mainly to Moody's becoming more favorable rather than S&P becoming tougher.
- Moody's post-IPO relative generosity is stronger for structured-product issuers and financial firms.
- It is also stronger for large/frequent bond issuers, marginal issuers, and industries with more Fitch competition.
- The same post-IPO pattern appears for outstanding bonds.

- The more favorable Moody's ratings are not more informative about expected default frequencies or actual defaults.
- Recovery rates do not rise enough after the IPO to justify more favorable Moody's ratings.
- Fitch's expansion does not fully explain the results.
- The CDO sample also shows Moody's post-IPO ratings becoming relatively more favorable.
- Robustness tests using alternative windows, Fitch benchmarks, firm/industry controls, and cross-sectional regressions support the main result.
- The central economic lesson is that public ownership can strengthen revenue incentives for gatekeepers whose quality is hard to observe immediately.

## 7 Conclusion

### 7.1 Core conclusion

Kedia, Rajgopal, and Zhou find that Moody's ratings became significantly more favorable to issuers relative to S&P after Moody's went public in 2000. The change is visible in new corporate bonds, outstanding bonds, and structured products. The evidence is strongest for clients and settings where conflicts of interest are likely to be more severe.

### 7.2 Economic intuition

Before the IPO, Moody's had stronger incentives to preserve a conservative reputation. After becoming public, measured performance, revenue, market share, and stock price became more salient. Since rating quality is hard to evaluate in real time but revenue is easy to measure, stronger market incentives may have encouraged leniency toward important clients.

### 7.3 Takeaway

The paper is a warning about the ownership structure of financial gatekeepers. Public ownership can improve focus and market discipline, but it can also create pressure to satisfy clients and shareholders in ways that weaken gatekeeping quality. In credit ratings, where mistakes may only be revealed years later, these incentives can have large consequences for capital markets.